

Question			Marking Details	Marks Available					
				AO1	AO2	AO3	TOTAL	Maths	Prac
3	(a)		Accept any answer in the range 613 000 - 624 000 = 3 marks Accept in standard form If incorrect award 2 marks for either $100/(0.09 \times 0.09) \times 3.14 (\pi) \times 156$ $100/0.025(43) \times 156$ If incorrect award 1 mark for area calculation $0.09 \times 0.09 \times 3.14 (\pi)$		3		3	3	
	(b)	i	Serial dilution description (of tenfold/ hundredfold) e.g. 1 cm ³ of sample and 9 cm ³ of {water/ growth medium} mixed (gives 10 ⁻¹) (1) or equivalent (does not have to give 10 ⁻⁵) Water/ saline/ equipment must be sterile / reference to aseptic technique(1) Repeated four more times/ method to achieve a 10 ⁻⁵ dilution (1) Accept annotated diagram	3			3		3
		ii	Plate 5 (1) (Plate 1/2/3/4)- cannot distinguish individual <u>colonies</u> / too many <u>colonies</u> to count accurately /merged <u>colonies</u> (might provide underestimate) (1) (Plate 6) has too few colonies to provide an accurate estimate (closer to the true value)/ too few to be statistically significant (1) Penalise bacteria once only			3	3		3
		iii	37°C is body temperature which is optimum for these bacteria/ 37°C grows the bacteria found in the patient / 37°C grows the bacteria found in the patient <u>faster</u> (1)		1		1		1
		iv	<u>Thick</u> {murein/peptidoglycan} layer (1) No lipopolysaccharide layer (1)	2			2		
			QUESTION 3 TOTAL	5	4	3	12	3	7